1.4 Abstract Algebra I: Study Notes

These notes condense a first course in groups and rings. Proofs are given in full where they are short and central, and sketched where the idea matters more than the details. Notation: e or 1 is the identity of a group G; $|G|$ is its order; $|g|$ is the order of the element g; $H \le G$ means H is a subgroup; $N \trianglelefteq G$ means N is a normal subgroup.

Part I: Groups

1. Groups and subgroups

A *group* is a set G with an associative binary operation, an identity element and inverses. It is *abelian* if the operation is commutative.

Standard examples

- $(\mathbb{Z}, +)$ and $(\mathbb{Z}_n, +)$, which are cyclic.
- $U(n) = \{[a] \in \mathbb{Z}_n : \gcd(a, n) = 1\}$ under multiplication, of order $\varphi(n)$.
- D_n, the symmetries of a regular n-gon, of order $2n$: $D_n = \langle r, s \mid r^n = s^2 = e,\ srs^{-1} = r^{-1} \rangle$.
- S_n, all permutations of $\{1, \dots, n\}$, of order $n!$.
- $GL_n(\mathbb{F})$ (invertible matrices), $SL_n(\mathbb{F})$ (determinant 1), $O(n)$ and $U(n)$.
- $Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}$ with $i^2 = j^2 = k^2 = ijk = -1$.

Subgroup test. A nonempty subset $H \subseteq G$ is a subgroup iff $a, b \in H \Rightarrow ab^{-1} \in H$. For finite H, closure under the operation is enough.

Order. $|g|$ is the least $n \ge 1$ with $g^n = e$, or ∞ if there is none. If $|g| = n$, then $g^k = e \iff n \mid k$, and $|g^k| = n / \gcd(n, k)$.

Pitfalls.

- $(ab)^{-1} = b^{-1}a^{-1}$, not $a^{-1}b^{-1}$.
- $|ab|$ need not relate to $|a|$ and $|b|$ in a non-abelian group. In S_3, two elements of order 2 have a product of order 3.

2. Cyclic groups

$\langle g \rangle = \{g^k : k \in \mathbb{Z}\}$. A group is *cyclic* if $G = \langle g \rangle$ for some g. Every cyclic group is isomorphic to $\mathbb{Z}$ or to some $\mathbb{Z}_n$.

Theorem. Every subgroup of a cyclic group is cyclic. If $|G| = n$ and G is cyclic, then for each $d \mid n$ there is **exactly one** subgroup of order d, namely $\langle g^{n/d} \rangle$. The generators of $\mathbb{Z}_n$ are the k with $\gcd(k, n) = 1$.

Proof that subgroups are cyclic. Let $H \le \langle g \rangle$ with $H \neq \{e\}$. Let m be the least positive integer with $g^m \in H$. If $g^k \in H$, write $k = qm + r$ with $0 \le r < m$. Then $g^r = g^k (g^m)^{-q} \in H$, so $r = 0$ by minimality. Hence $H = \langle g^m \rangle$. $\square$

3. Permutation groups

Every permutation is a product of disjoint cycles, uniquely up to order. Its *cycle type* is the multiset of cycle lengths. The order of a permutation is the lcm of its cycle lengths.

Every permutation is also a product of transpositions. The parity of the number of transpositions is well defined, which gives the **sign** homomorphism $\operatorname{sgn} : S_n \to \{\pm 1\}$. A k-cycle has sign $(-1)^{k-1}$. The kernel of sgn is the alternating group A_n, of order $n!/2$ for $n \geq 2$.

Conjugation in S_n.

$$\sigma(a_1\, a_2\, \ldots\, a_k)\sigma^{-1} = (\sigma(a_1)\, \sigma(a_2)\, \ldots\, \sigma(a_k)).$$

So two permutations are conjugate in S_n iff they have the same cycle type. Conjugacy classes of S_n correspond to partitions of n. In A_n a class of S_n can split into two; see the A_5 computation in Section 9.

4. Cosets and Lagrange's theorem

For $H \leq G$, the *left cosets* gH partition G, because $aH = bH \iff a^{-1}b \in H$. Each coset has $|H|$ elements, since $h \mapsto gh$ is a bijection $H \to gH$.

Lagrange's Theorem. If G is finite and $H \leq G$, then $|H|$ divides $|G|$, and $[G : H] = |G|/|H|$ is the number of cosets.

Corollaries.

- $|g|$ divides $|G|$, so $g^{|G|} = e$.
- A group of prime order is cyclic, generated by any non-identity element.
- **Euler:** $a^{\varphi(n)} \equiv 1 \pmod{n}$ if $\gcd(a, n) = 1$. **Fermat:** $a^{p-1} \equiv 1 \pmod{p}$ if $p \nmid a$.

The converse of Lagrange is false. A_4 has order 12 but no subgroup of order 6 (Problem 14). Partial converses are given by Cauchy's theorem (Section 8) and Sylow's theorems (Section 9).

Product formula. For subgroups H, K, $|HK| = |H||K|/|H \cap K|$. Here HK need not be a subgroup. It is one iff $HK = KH$, which holds for example when one of H, K is normal.

5. Homomorphisms, normal subgroups and quotients

A *homomorphism* is a map $\varphi : G \to H$ with $\varphi(ab) = \varphi(a)\varphi(b)$. It follows that $\varphi(e) = e$ and $\varphi(a^{-1}) = \varphi(a)^{-1}$. The *kernel* $\ker\varphi$ is a subgroup, and φ is injective iff $\ker\varphi = \{e\}$.

Normal subgroups. $N \leq G$ is *normal* if $gNg^{-1} = N$ for all g. Equivalent conditions:

- $gNg^{-1} \subseteq N$ for all g;
- $gN = Ng$ for all g;
- N is a union of conjugacy classes;
- N is the kernel of some homomorphism.

Kernels are normal: if $\varphi(n) = e$, then $\varphi(gng^{-1}) = \varphi(g)\varphi(g)^{-1} = e$. Every subgroup of an abelian group is normal. Every subgroup of index 2 is normal (Problem 9).

Quotient group. If $N \trianglelefteq G$, the cosets form a group G/N with $(aN)(bN) = abN$. Normality is **exactly** what makes this well defined. If $aN = a'N$ and $bN = b'N$, write $a' = an_1$ and $b' = bn_2$. Then $a'b' = an_1bn_2 = ab(b^{-1}n_1b)n_2 \in abN$. The map $\pi : G \to G/N$ is a surjective homomorphism with kernel N.

First Isomorphism Theorem. If $\varphi : G \to H$ is a homomorphism, then $G/\ker\varphi \cong \operatorname{im}\varphi$, via $g\ker\varphi \mapsto \varphi(g)$.

Proof. The map is well defined and injective because $\varphi(a) = \varphi(b) \iff a^{-1}b \in \ker\varphi \iff a\ker\varphi = b\ker\varphi$. It is a homomorphism because φ is, and it is clearly onto $\operatorname{im}\varphi$. □

How to use it. To identify G/N, find a surjective homomorphism out of G whose kernel is N. Examples:

- $\mathbb{Z}/n\mathbb{Z} \cong \mathbb{Z}_n$.
- $GL_n/SL_n \cong \mathbb{F}^\times$, via det.
- $\mathbb{R}/\mathbb{Z} \cong S^1$, via $t \mapsto e^{2\pi i t}$.
- $S_n/A_n \cong \{\pm 1\}$.

Second (diamond) Isomorphism Theorem. If $H \leq G$ and $N \trianglelefteq G$, then $HN \leq G$, $H \cap N \trianglelefteq H$, and $HN/N \cong H/(H \cap N)$.

Third Isomorphism Theorem. If $N \leq K$ are both normal in G, then $(G/N)/(K/N) \cong G/K$.

Correspondence Theorem. For $N \trianglelefteq G$, subgroups of G/N correspond bijectively to subgroups of G containing N, via $K \mapsto K/N$. The correspondence preserves inclusion, index and normality.

Simple groups. $G \neq \{e\}$ is *simple* if its only normal subgroups are $\{e\}$ and G. The abelian simple groups are exactly the $\mathbb{Z}_p$. Simple groups are the "atoms" from which all finite groups are built (Jordan–Hölder).

6. Direct products and finite abelian groups

$G \times H$ has the componentwise operation, and $|(g,h)| = \text{lcm}(|g|,|h|)$. Hence $\mathbb{Z}_m \times \mathbb{Z}_n \cong \mathbb{Z}_{mn}$ iff $\gcd(m,n) = 1$.

Internal direct product. If $H, K \trianglelefteq G$, $H \cap K = \{e\}$ and $HK = G$, then $G \cong H \times K$. The key step: for $h \in H$ and $k \in K$, the commutator $hkh^{-1}k^{-1}$ lies in $H \cap K$, so $hk = kh$.

Fundamental Theorem of Finite Abelian Groups (statement). Every finite abelian group is isomorphic to a product $\mathbb{Z}_{p_1^{a_1}} \times \cdots \times \mathbb{Z}_{p_r^{a_r}}$ of cyclic groups of prime-power order (the primes need not be distinct), uniquely up to order. Equivalently, it is $\mathbb{Z}_{d_1} \times \cdots \times \mathbb{Z}_{d_s}$ with $d_1 \mid d_2 \mid \cdots \mid d_s$ (invariant factors).

Example. The abelian groups of order $72 = 2^3 \cdot 3^2$ correspond to a partition of 3 and a partition of 2, so there are $3 \times 2 = 6$ of them.

This theorem is a special case of the structure theorem for finitely generated modules over a PID, which is proved in 2.3/3.3. The same theorem gives the Jordan form from 1.3.

7. Group actions

An *action* of G on a set X is a map $G \times X \to X$, $(g,x) \mapsto g \cdot x$, with $e \cdot x = x$ and $g \cdot (h \cdot x) = (gh) \cdot x$. Equivalently, it is a homomorphism $G \to \text{Sym}(X)$.

- The *orbit* of x is $\mathcal{O}_x = G \cdot x$. The orbits partition X.
- The *stabiliser* of x is $G_x = \{g : g \cdot x = x\} \leq G$.

Orbit–Stabiliser Theorem. The map $gG_x \mapsto g \cdot x$ is a well-defined bijection from G/G_x (the set of cosets) to $\mathcal{O}_x$. So if G is finite, $|\mathcal{O}_x| = [G : G_x]$ and

$$|G| = |\mathcal{O}_x| \cdot |G_x|.$$

Example. The rotation group of a cube acts transitively on the 6 faces. The stabiliser of a face is the 4 rotations about the axis through that face's centre. So the group has order 24. It is isomorphic to S_4, acting on the 4 long diagonals.

Key actions.

1. **Left multiplication** on G itself. This gives **Cayley's Theorem**: every group embeds in $\text{Sym}(G)$, so every group of order n embeds in S_n.

2. **Left multiplication on cosets** G/H. This gives a homomorphism $G \to S_{[G:H]}$ whose kernel lies in H. *Useful consequence:* if G is simple and $[G : H] = k > 1$, then G embeds in S_k, so $|G|$ divides $k!$.
3. **Conjugation** $g \cdot x = gxg^{-1}$. The orbits are conjugacy classes. The stabiliser of x is the centraliser $C_G(x)$. The fixed points are the centre $Z(G)$.

Class Equation. If G is finite and $x_1, \dots, x_r$ are representatives of the conjugacy classes with more than one element, then

$$|G| = |Z(G)| + \sum_{i=1}^{r} [G : C_G(x_i)].$$

Corollary. A nontrivial finite p-group (order p^n, $n \geq 1$) has nontrivial centre. *Proof.* Each $[G : C_G(x_i)]$ is a power of p greater than 1, so it is divisible by p. Hence p divides $|Z(G)|$, and $|Z(G)| \geq p$. □ Consequence: every group of order p^2 is abelian (Problem 13).

Burnside's counting lemma. The number of orbits equals $\frac{1}{|G|} \sum_{g \in G} |\mathrm{Fix}(g)|$. It is the standard tool for counting colourings up to symmetry.

8. Cauchy's theorem

Cauchy. If a prime p divides $|G|$, then G has an element of order p.

Slick proof (McKay). Let $X = \{(g_1, \dots, g_p) \in G^p : g_1 g_2 \cdots g_p = e\}$. Then $|X| = |G|^{p-1}$, because the first $p - 1$ entries are free and determine the last. $\mathbb{Z}_p$ acts on X by cyclic shifts; a shift of a solution is still a solution, since $ab = e \Rightarrow ba = e$. Orbits have size 1 or p. Since $p \mid |X|$, the number of fixed points is divisible by p. The fixed points are the tuples $(g, \dots, g)$ with $g^p = e$. One of them is $(e, \dots, e)$, so there are at least $p - 1 \geq 1$ others, and each gives an element of order p. □

9. Sylow theorems

Write $|G| = p^a m$ with $p \nmid m$. A *Sylow p-subgroup* is a subgroup of order p^a. Let n_p be the number of Sylow p-subgroups.

Sylow I (existence). Sylow p-subgroups exist. More generally, G has a subgroup of every order p^k with $k \leq a$. **Sylow II (conjugacy).** Any two Sylow p-subgroups are conjugate. Every p-subgroup is contained in some Sylow p-subgroup. **Sylow III (counting).** $n_p \equiv 1 \pmod{p}$, $n_p \mid m$, and $n_p = [G : N_G(P)]$ for any Sylow p-subgroup P.

Proof ideas.

- I: either Wielandt's argument (let G act on the set of all p^a-element subsets and count mod p), or induction on $|G|$ using the class equation and Cauchy.
- II: let a p-subgroup Q act on the cosets G/P. The number of cosets $[G : P] = m$ is prime to p, so the action has a fixed point gP. That means $Q \subseteq gPg^{-1}$.
- III: $n_p \mid m$ because $n_p = [G : N_G(P)]$ by orbit–stabiliser for the conjugation action, and $P \leq N_G(P)$. For $n_p \equiv 1 \pmod{p}$, let P act on the set of Sylow subgroups by conjugation. P is the only fixed point, and all other orbits have size divisible by p.

Key consequence. $n_p = 1$ iff the Sylow p-subgroup is normal.

Standard applications.

- $|G| = pq$ *with* $p < q$ *primes and* $p \nmid q-1 \Rightarrow G \cong \mathbb{Z}_{pq}$. Examples: 15, 33, 35. *Proof:* $n_q \mid p$ and $n_q \equiv 1 \pmod{q}$ force $n_q = 1$. $n_p \mid q$ and $n_p \equiv 1 \pmod{p}$ force $n_p = 1$, since $q \not\equiv 1 \pmod{p}$. So both Sylow subgroups are normal, they intersect trivially, and $G \cong \mathbb{Z}_p \times \mathbb{Z}_q \cong \mathbb{Z}_{pq}$.
- *No simple group of order* n. Find a prime p with n_p forced to be 1. If that fails, count elements: distinct subgroups of prime order p intersect trivially, so n_p of them contribute $n_p(p-1)$ elements of order p. If these counts exceed n, some $n_p = 1$. Orders 12, 20, 30, 56 and 63 are all handled this way.
- A_5 **is simple.** Its class sizes are $1, 15, 20, 12, 12$. A normal subgroup is a union of classes containing $\{e\}$ whose size divides 60, and only 1 and 60 qualify. More generally, A_n is simple for $n \geq 5$. A_5 (order 60) is the smallest non-abelian simple group.

Part II: Rings

10. Rings, ideals and quotients

A *ring* R has addition (an abelian group) and an associative multiplication that distributes over addition. In this course **rings have a** 1 and are usually commutative. A *unit* is an element with a multiplicative inverse; the units form a group $R^\times$. A *zero divisor* is $a \neq 0$ with $ab = 0$ for some $b \neq 0$.

- An *integral domain* is a commutative ring with $1 \neq 0$ and no zero divisors.
- A *field* is a commutative ring with $1 \neq 0$ in which every nonzero element is a unit.
- Every finite integral domain is a field: $x \mapsto ax$ is injective, hence surjective.

Examples. $\mathbb{Z}$; $\mathbb{Z}_n$ (a field iff n is prime); $\mathbb{Q}, \mathbb{R}, \mathbb{C}$; $R[x]$; $M_n(R)$ (non-commutative); $\mathbb{Z}[i]$; $C[0,1]$ (has zero divisors); $\mathbb{Z}[\sqrt{-5}]$.

Ideals. An *ideal* $I \subseteq R$ is an additive subgroup with $rI \subseteq I$ for all $r \in R$. Ideals are exactly the kernels of ring homomorphisms. If an ideal contains a unit, it is all of R. Notation: $(a) = aR$ is a principal ideal, and $(a_1, \dots, a_n)$ is the ideal generated by $a_1, \dots, a_n$.

Quotient ring. R/I consists of the additive cosets $a+I$, with $(a+I)(b+I) = ab+I$. This is well defined *because* I absorbs multiplication. The isomorphism theorems and the correspondence theorem hold just as for groups. In particular $R/\ker\varphi \cong \operatorname{im}\varphi$.

Characteristic. $\operatorname{char} R$ is the least $n \geq 1$ with $n \cdot 1 = 0$, or 0 if there is none. The characteristic of an integral domain is 0 or a prime.

11. Prime and maximal ideals

Let R be commutative. An ideal $P \neq R$ is *prime* if $ab \in P \Rightarrow a \in P$ or $b \in P$. An ideal $M \neq R$ is *maximal* if there is no ideal strictly between M and R.

Theorem.

- R/P is an integral domain iff P is prime.
- R/M is a field iff M is maximal.
- Hence every maximal ideal is prime.

Proof of the field statement. By the correspondence theorem, the ideals of R/M correspond to ideals of R containing M. A commutative ring with $1 \neq 0$ is a field iff its only ideals are 0 and itself. □

Examples in $\mathbb{Z}[x]$**.**

- (x) is prime but not maximal: $\mathbb{Z}[x]/(x) \cong \mathbb{Z}$.

- $(2, x)$ is maximal: $\mathbb{Z}[x]/(2, x) \cong \mathbb{F}_2$.
- $(x^2 + 1)$ is prime, since $\mathbb{Z}[x]/(x^2 + 1) \cong \mathbb{Z}[i]$.

Field of fractions. Every integral domain D embeds in a smallest field $\mathrm{Frac}(D)$, consisting of formal fractions a/b with $a/b = c/d \iff ad = bc$. For example, $\mathrm{Frac}(\mathbb{Z}) = \mathbb{Q}$ and $\mathrm{Frac}(F[x]) = F(x)$.

Chinese Remainder Theorem. If $I + J = R$ (the ideals are *comaximal*), then $R/(I \cap J) \cong R/I \times R/J$, and $I \cap J = IJ$ when R is commutative. For $R = \mathbb{Z}$: $\mathbb{Z}_{mn} \cong \mathbb{Z}_m \times \mathbb{Z}_n$ as rings when $\gcd(m, n) = 1$. Taking units gives $U(mn) \cong U(m) \times U(n)$, so φ is multiplicative.

12. Factorisation: Euclidean domains, PIDs and UFDs

In an integral domain D:

- a and b are *associates* if $a = ub$ for a unit u.
- A non-unit $p \neq 0$ is *irreducible* if $p = ab$ implies a or b is a unit.
- A non-unit $p \neq 0$ is *prime* if $p \mid ab$ implies $p \mid a$ or $p \mid b$.
- Prime ⇒ irreducible always. The converse holds in UFDs but can fail in general.

Definitions.

- *Euclidean domain:* there is a norm $N : D \setminus \{0\} \to \mathbb{Z}_{\geq 0}$ with a division algorithm. For all a and all $b \neq 0$, there are q, r with $a = qb + r$ and either $r = 0$ or $N(r) < N(b)$.
- *PID:* an integral domain in which every ideal is principal.
- *UFD:* every nonzero non-unit is a product of irreducibles, uniquely up to order and associates.

Hierarchy.

$$\text{fields} \subset \text{Euclidean domains} \subset \text{PIDs} \subset \text{UFDs} \subset \text{integral domains}.$$

- **ED ⇒ PID.** Given an ideal $I \neq 0$, pick $b \in I$ of minimal norm. For $a \in I$, divide: $a = qb + r$. Then $r = a - qb \in I$ has smaller norm than b, so $r = 0$. Hence $I = (b)$.
- **PID ⇒ UFD.** Existence: in a PID, ascending chains of ideals stabilise (the union of a chain is an ideal, hence principal, and its generator lies in some member of the chain). Uniqueness: in a PID, irreducibles are prime. If p is irreducible, (p) is maximal among principal ideals, hence maximal, hence prime.
- **In a PID, every nonzero prime ideal is maximal** (Problem 22).

Examples and counterexamples.

Ring	ED?	PID?	UFD?
$\mathbb{Z}$, $F[x]$, $\mathbb{Z}[i]$	yes	yes	yes
$\mathbb{Z}[\frac{1+\sqrt{-19}}{2}]$	no	yes	yes
$\mathbb{Z}[x]$, $F[x, y]$	no	no	yes
$\mathbb{Z}[\sqrt{-5}]$	no	no	no

- $\mathbb{Z}[i]$ is Euclidean with $N(a + bi) = a^2 + b^2$. To divide, round the complex quotient to the nearest Gaussian integer; the error has norm at most $1/2$.
- $\mathbb{Z}[x]$ is a UFD by Gauss's lemma, but $(2, x)$ is not principal.
- In $\mathbb{Z}[\sqrt{-5}]$, $6 = 2 \cdot 3 = (1 + \sqrt{-5})(1 - \sqrt{-5})$ gives two essentially different factorisations into irreducibles.
- The $\mathbb{Z}[\frac{1+\sqrt{-19}}{2}]$ example is standard but hard to verify (D&F 8.2); treat it as a known fact.

Fact. If R is a UFD, then $R[x]$ is a UFD. Hence $\mathbb{Z}[x_1, \dots, x_n]$ and $F[x_1, \dots, x_n]$ are UFDs.

Pitfall. Gcds exist in any UFD, but **Bézout** ($\gcd(a,b) = ra + sb$) needs a PID. In $\mathbb{Z}[x]$, $\gcd(2, x) = 1$, yet $1 \notin (2, x)$.

13. Polynomial rings

Division algorithm. Let F be a field, $f, g \in F[x]$ and $g \neq 0$. There are unique q, r with $f = qg + r$ and $\deg r < \deg g$. So $F[x]$ is a Euclidean domain (norm = degree), hence a PID and a UFD.

Roots. a is a root of f iff $(x - a) \mid f$. A nonzero polynomial of degree n over an integral domain has at most n roots. *Consequence:* any finite subgroup of $F^\times$ is cyclic. *Sketch:* if it were not cyclic, its exponent m would be smaller than its order, but then $x^m - 1$ would have more than m roots. In particular, $\mathbb{F}_q^\times$ is cyclic.

Irreducibility tests over $\mathbb{Q}$.

1. *Degree 2 or 3:* irreducible iff there is no root in F. This fails for degree 4: $(x^2 + 1)^2$ has no rational root but is reducible.
2. *Rational root test:* a root r/s in lowest terms of $a_n x^n + \cdots + a_0 \in \mathbb{Z}[x]$ has $r \mid a_0$ and $s \mid a_n$.
3. **Gauss's lemma:** a primitive $f \in \mathbb{Z}[x]$ is irreducible in $\mathbb{Z}[x]$ iff it is irreducible in $\mathbb{Q}[x]$. Equivalently, if f factors over $\mathbb{Q}$, it factors over $\mathbb{Z}$ with factors of the same degrees.
4. **Eisenstein:** if a prime p divides every coefficient except the leading one, $p \nmid a_n$, and $p^2 \nmid a_0$, then f is irreducible over $\mathbb{Q}$. A substitution $x \mapsto x + 1$ often helps: $\Phi_p(x) = \frac{x^p - 1}{x - 1}$ becomes Eisenstein at p.
5. *Reduction mod p:* if $p \nmid a_n$ and $\bar{f}$ is irreducible in $\mathbb{F}_p[x]$, then f is irreducible over $\mathbb{Q}$. The converse fails: $x^4 + 1$ is irreducible over $\mathbb{Q}$ but reducible mod every prime.

Constructing fields. If $f \in F[x]$ is irreducible of degree n, then (f) is maximal, and $K = F[x]/(f)$ is a field. It is an n-dimensional vector space over F with basis $1, \alpha, \dots, \alpha^{n-1}$, where $\alpha = x + (f)$ is a root of f. This is **Kronecker's construction**, and it is how every finite field arises: $\mathbb{F}_{p^n} = \mathbb{F}_p[x]/(f)$ for any irreducible f of degree n.

Examples.

- $\mathbb{R}[x]/(x^2 + 1) \cong \mathbb{C}$.
- $\mathbb{Q}[x]/(x^3 - 2) \cong \mathbb{Q}(\sqrt[3]{2})$.
- $\mathbb{F}_2[x]/(x^2 + x + 1) = \mathbb{F}_4$.

Common pitfalls (summary)

1. Assuming the converse of Lagrange. A_4 has no subgroup of order 6.
2. Defining a quotient G/H when H is not normal. The coset multiplication is then not well defined.
3. Forgetting to check that a map out of a quotient is *well defined*.
4. Concluding G is not simple because $n_p > 1$. You need some $n_p = 1$, or another argument such as counting elements or embedding in S_k.
5. Mixing up irreducible and prime elements. They coincide in UFDs, not in general.
6. Applying Eisenstein to a non-primitive polynomial, or forgetting the $p^2 \nmid a_0$ condition.
7. "No roots ⇒ irreducible" only holds in degree 2 or 3.

Connections forward

- **2.3 Abstract Algebra II:** field extensions, splitting fields, Galois groups. The Galois correspondence is a correspondence theorem between subgroups and intermediate fields. Solvability of groups decides solvability by radicals. Modules over PIDs give both the structure theorem for abelian groups and the Jordan form.
- **3.3 Graduate Algebra:** Noetherian rings, localisation, tensor products, representation theory of finite groups (built on conjugacy classes and group actions), and homological algebra.
- **3.4 Algebraic Topology:** the fundamental group. Covering spaces correspond to subgroups of π_1 (another Galois-type correspondence), and deck transformations are a group action. Homology groups are abelian groups computed as quotients, and the structure theorem describes them.
- **3.5 Smooth Manifolds:** Lie groups such as GL_n, $O(n)$ and $SU(2)$ act smoothly on manifolds, and quotients by group actions give new manifolds.
- **3.1 / 3.2 Analysis:** quotient constructions (L^p, quotient Banach spaces) follow the same "kernel of a homomorphism" pattern. In Banach algebras, maximal ideals correspond to characters (Gelfand theory).